

A Well Routed Router.

But ? What is a router ?!

A router is a Layer 3 network device that performs packet forwarding between distinct IP subnets by consulting its routing table and selecting the most appropriate path according to metrics, administrative distance, and protocol-learned reachability.

It operates as the default gateway for end hosts, encapsulating and transmitting traffic toward remote networks while maintaining broadcast domain separation and enabling inter-network communication. Routers can use static routes or dynamic routing protocols such as OSPF, EIGRP, or BGP to determine path selection.

In enterprise environments, routers also provide traffic engineering and policy enforcement through mechanisms such as NAT, ACLs, QoS, route redistribution, VPN termination, and control-plane security.

Know Before configuring a Cisco router, the key points to consider are:

1- Define the router's role

- Determine the business/technical purpose: internet edge, WAN branch, site-to-site, core/internal routing, VPN hub, etc.
- Estimate how many interfaces, subnets, and users it must support.
- Identify performance and availability expectations.

2- Prepare the IP addressing plan

- Interface IP addresses and subnet masks
- Default gateways and next hops
- VLAN IDs and subnet-to-VLAN mapping (if applicable)
- Reserved ranges (management, servers, voice, guest, etc.)

3- Understand the topology

- What devices are connected to each interface
- Uplink/downlink/WAN mapping
- Redundant links and failover design
- Bandwidth and latency constraints

4- Choose the routing strategy

- Static routing for small, stable networks
- Dynamic routing (OSPF/EIGRP/BGP/RIP) for scalable or changing environments
- Plan route summarization, default-route propagation, and route filtering policies

5- Secure the router from day one

- Set strong local credentials and enable secret
- Use SSH only (disable Telnet)
- Configure ACLs for management and traffic control
- Add a legal/security banner
- Disable unused services and interfaces
- Consider control-plane security (CoPP) and login protections

6- Verify hardware/software readiness

- Router model and interface types
- IOS/IOS-XE version compatibility
- Available RAM/flash and storage
- Installed modules/licenses and feature availability

7- Prepare backup and recovery

- Save baseline configs before changes
- Define rollback procedure
- Back up startup/running configs and IOS images
- Document password recovery and disaster recovery steps

9- Identify required network services

- NAT/PAT
- DHCP server or relay
- Inter-VLAN routing
- VPN (site-to-site/remote access)
- QoS policies
- ACL/firewall-like filtering
- High availability (HSRP/VRRP/GLBP, dual WAN, etc.)

10- Define validation and acceptance tests

- Ping/traceroute end-to-end tests
- Routing table and neighbor validation
- NAT, ACL, and VPN functionality checks
- Failover and redundancy tests
- Remote management and security verification

11- Document everything

- Hostname and device role
- Interface descriptions and status
- IP addressing and VLAN table

- Routing protocols and key parameters
- Security policies (ACLs, management restrictions)
- Change log, test results, and final approved configuration

Basic Router Configuration Step-by-Step

0- Pre-checks (before configuration)

```
enable
show version
show ip interface brief
show inventory
show running-config
```

1- Enter configuration mode + basic identity

```
configure terminal
hostname R1
no ip domain-lookup
service password-encryption
enable secret <STRONG_SECRET>
banner motd "Unauthorized access is prohibited."
```

2- Secure management access (SSH only)

```
ip domain-name company.local
username admin privilege 15 secret <STRONG_PASSWORD>
crypto key generate rsa modulus 2048
ip ssh version 2
```

```
line console 0
logging synchronous
exec-timeout 10 0
login local
exit
```

```
line vty 0 4
transport input ssh
login local
exec-timeout 10 0
exit
```

3- Configure Layer-3 interfaces (or subinterfaces)

Case A: Routed physical interfaces

```
interface g0/0
description LAN_USERS
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface g0/1
description WAN_TO_ISP
ip address 203.0.113.2 255.255.255.252
no shutdown
exit
```

Case B: Router-on-a-stick (inter-VLAN routing)

```
interface g0/0
no shutdown
exit

interface g0/0.10
encapsulation dot1Q 10
description VLAN10_USERS
ip address 192.168.10.1 255.255.255.0
exit

interface g0/0.20
encapsulation dot1Q 20
description VLAN20_SERVERS
ip address 192.168.20.1 255.255.255.0
exit
```

4- Configure routing

Option A: Static routing

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip route 10.10.0.0 255.255.0.0 192.168.20.2
```

Option B: OSPF (single area example)

```
router ospf 1
router-id 1.1.1.1
```

```
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 203.0.113.0 0.0.0.3 area 0
passive-interface g0/0.10
passive-interface g0/0.20
exit
```

5- NAT/PAT (internet access for inside networks)

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
```

```
interface g0/0
ip nat inside
exit
```

```
interface g0/1
ip nat outside
exit
```

```
ip nat inside source list 1 interface g0/1 overload
```

6- ACL policy (traffic filtering)

```
ip access-list extended VLAN10_FILTER
deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
permit ip 192.168.10.0 0.0.0.255 any
exit
```

```
interface g0/0.10
ip access-group VLAN10_FILTER in
exit
```

7- Optional DHCP service (if router is DHCP server)

```
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp pool VLAN10_POOL
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8 1.1.1.1
lease 7
exit
```

8- Validation and troubleshooting commands

```
show ip interface brief
show interfaces description
show ip route
show ip ospf neighbor
show ip protocols
```

```
show access-lists
show ip nat translations
show ip nat statistics
show running-config | section nat
show running-config | section access-list
ping 8.8.8.8
traceroute 8.8.8.8
```

9- Save and backup

```
copy running-config startup-config
(Optional backup to server)
copy startup-config tftp:
copy running-config tftp:
```

Switch Configuration

But ? What is a switch ?!

A switch operates mainly at Layer 2 of the OSI model (the Data Link layer). It learns the MAC address of each connected device and stores those addresses in a MAC address table. When it receives an Ethernet frame, it examines the destination MAC address and sends the frame only to the port associated with that destination, instead of sending it to every port.

1- Enter privileged mode, global config, and set hostname

```
enable
configure terminal
hostname SW1
```

2- Set privileged EXEC password

```
enable secret <STRONG_SECRET>
```

3- Secure console access

```
line console 0
password <CONSOLE_PASSWORD>
login
exec-timeout 10 0
logging synchronous
exit
```

4- Configure remote access (prefer SSH, avoid Telnet)

```
Minimum legacy VTY password (not recommended alone)
line vty 0 15
password <VTY_PASSWORD>
login
exit
```

Recommended secure method (SSH)

```
ip domain-name company.local
username admin privilege 15 secret <ADMIN_PASSWORD>
crypto key generate rsa modulus 2048
ip ssh version 2

line vty 0 15
transport input ssh
login local
exec-timeout 10 0
exit
service password-encryption
```

5- Configure MOTD security banner

```
banner motd "Unauthorized access is prohibited."
```

VLANs Configuration

Subnet Name	Needed Hosts	Subnet Address	Gateway IP	IP Range	Subnet Mask	Usable IPs	VLAN ID
IT Department	5	192.168.10.0/24	192.168.10.254	192.168.10.1 - 192.168.10.253	255.255.255.0	254	10
Production	10	192.168.20.0/24	192.168.20.254	192.168.20.1 - 192.168.20.253	255.255.255.0	254	20
Management /Secretariat	6	192.168.30.0/24	192.168.30.254	192.168.30.1 - 192.168.30.253	255.255.255.0	254	30
Support Sector 1	11	192.168.40.0/24	192.168.40.254	192.168.40.1 - 192.168.40.253	255.255.255.0	254	40
Support Sector 2	11	192.168.50.0/24	192.168.50.254	192.168.50.1 - 192.168.50.253	255.255.255.0	254	50
Study/R&D	9	192.168.60.0/24	192.168.60.254	192.168.60.1 - 192.168.60.253	255.255.255.0	254	60
Servers	6	192.168.70.0/24	192.168.70.254	192.168.70.1 - 192.168.70.253	255.255.255.0	254	70
DMZ	2	192.168.80.0/24	192.168.80.254	192.168.80.1 - 192.168.80.253	255.255.255.0	254	80

Cisco commands

1- Create VLANs on switch

```
enable
configure terminal

vlan 10
name IT
exit

vlan 20
name PRODUCTION
exit

vlan 30
name MANAGEMENT
exit

vlan 40
name SUPPORT1
exit

vlan 50
name SUPPORT2
exit

vlan 60
name RD
exit

vlan 70
name SERVERS
exit

vlan 80
name DMZ
exit
```

2- Affect access ports for VLANs

! VLAN 10

```
interface range fastethernet0/1 - 5
switchport mode access
switchport access vlan 10
spanning-tree portfast
exit
```

! VLAN 20

```
interface range fastethernet0/6 - 15
switchport mode access
switchport access vlan 20
spanning-tree portfast
exit
```

! VLAN 30

```
interface range fastethernet0/16 - 20
switchport mode access
switchport access vlan 30
spanning-tree portfast
exit
```

! VLAN 40

```
interface range fastethernet0/21 - 24
switchport mode access
switchport access vlan 40
spanning-tree portfast
exit
```

3- Configure trunk link (switch ↔ router or switch ↔ L3 switch)

```
interface gigabitethernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
```

Optional hardening (recommended):

```
interface gigabitethernet0/1
switchport trunk native vlan 999
```

4- Inter-VLAN routing (choose ONE method)

Method A — Router-on-a-Stick (router does Layer 3)

Use this when your switch is Layer 2

On the switch (trunk to router)

```
interface gigabitethernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80
no shutdown
exit
```

On the router (subinterfaces)

enable
configure terminal

interface g0/0
no shutdown
exit

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.1.65 255.255.255.248
exit

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.1.1 255.255.255.240
exit

interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.1.73 255.255.255.248
exit

interface g0/0.40
encapsulation dot1Q 40
ip address 192.168.1.17 255.255.255.240
exit

interface g0/0.50
encapsulation dot1Q 50
ip address 192.168.1.33 255.255.255.240
exit

interface g0/0.60
encapsulation dot1Q 60
ip address 192.168.1.49 255.255.255.240
exit

interface g0/0.70
encapsulation dot1Q 70
ip address 192.168.1.81 255.255.255.248
exit

interface g0/0.80
encapsulation dot1Q 80
ip address 192.168.1.89 255.255.255.252
exit

Method B — Layer 3 switch SVIs (switch does Layer 3)

Use this when your switch supports ip routing.

```
enable
configure terminal
ip routing

interface vlan 10
ip address 192.168.1.65 255.255.255.248
no shutdown
exit

interface vlan 20
ip address 192.168.1.1 255.255.255.240
no shutdown
exit

interface vlan 30
ip address 192.168.1.73 255.255.255.248
no shutdown
exit

interface vlan 40
ip address 192.168.1.17 255.255.255.240
no shutdown
exit

interface vlan 50
ip address 192.168.1.33 255.255.255.240
no shutdown
exit

interface vlan 60
ip address 192.168.1.49 255.255.255.240
no shutdown
exit

interface vlan 70
ip address 192.168.1.81 255.255.255.248
no shutdown
exit

interface vlan 80
ip address 192.168.1.89 255.255.255.252
no shutdown
exit
```

5) Configure OSPF (only if dynamic routing is needed)

```
router ospf 1
router-id 2.2.2.2
network 192.168.1.64 0.0.0.7 area 0
network 192.168.1.0 0.0.0.15 area 0
network 192.168.1.72 0.0.0.7 area 0
network 192.168.1.16 0.0.0.15 area 0
network 192.168.1.32 0.0.0.15 area 0
network 192.168.1.48 0.0.0.15 area 0
network 192.168.1.80 0.0.0.7 area 0
network 192.168.1.88 0.0.0.3 area 0
exit
```

6) Validation commands (very important)

```
show vlan brief
show interfaces trunk
show ip interface brief
show ip route
show ip ospf neighbor
show spanning-tree interface fastethernet0/1 detail
ping <gateway-or-remote-ip>
traceroute <remote-ip>
```

7) Save configuration

```
end
write memory
```